CS 18200 ✧ Spring 2025
Homework 01

Due: Friday, January 31, 2025 at 11:00 pm Eastern Time

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- Homework **`must be typed`** and uploaded as one .pdf file to Gradescope.
- Every problem has to start on a new page.
- Follow Gradescope instructions on matching problems to page numbers.
- Homework not following guidelines will not be graded.
- Answering to Problem 1 is **`mandatory`** for everyone. Failure to submit a solution to Problem 1 results in student submission not being graded.

Reminder of CS182 Academic Integrity:

What you submit as your solution for homework problems and what is graded needs to be **your own work**. It needs to be expressed and explained in **your own words**.
The line between discussing a problem with other students and using an idea developed by others and expressing your solution in your own words can at times be blurred. In addition, using a similar or related solution or an idea found online has significant potential for submitting something as your own that is easily detected as being copied. To be able to better assess academic honesty, we expect all students to provide for **every problem** on **every homework** information on the use of online material.
The following are considered **`cheating`** on homework.

- Sharing or copying solutions is unacceptable and is considered cheating.
- When using material or adapting ideas found online, you are required to give credit and proper references. Failure to do so is considered cheating.
- Anyone working with a tutor on homework or a project must disclose this information. Failure to do so is considered cheating.
- Making an online post or asking someone to make a post on a site like StackExchange, Chegg, Coursehero, or Reddit to get hints is considered cheating.
- You are expected to take reasonable precautions to prevent others from using your work. Failure to do so is considered cheating.

Take a look at the course syllabus on Brightspace for penalties.

Problem 1. **(MANDATORY)** In this question you must write the following sentence with your full name typed in between:

> *I,* **[YOUR FULL NAME GOES HERE]***, affirm that I have not given or received any unauthorized help on this assignment and that this work is my own. What I have submitted is expressed and explained in my own words. I have not used any online websites that provide a solution. I will not post any parts of this problem set to any online platform and doing so is a violation of course policy.*

So for example, if "John Doe" is in the class, they will submit the following as the solution to this problem:

> *I, John Doe, affirm that I have not given or received any unauthorized help on this assignment and that this work is my own. What I have submitted is expressed and explained in my own words. I have not used any online websites that provide a solution. I will not post any parts of this problem set to any online platform and doing so is a violation of course policy.*

Problem 2. **(8×3 points)**
Prove if the following statements are a tautology (always True) or a contradiction (always False). For tautologies/contradictions, use equivalences to show it (state the names of the equivalences used in your proof). If it is not a tautology or contradiction, give a counterexample (i.e. show that the statement can be true or false when T or F is assigned to p and q). Do NOT use truth tables.

(a) $((p \vee (q \to \neg p)) \wedge (p \vee (\neg q \to p))) \vee (p \to q)$

(b) $\neg(\neg q \vee (\neg(\neg q \wedge p) \wedge q)) \wedge p$

(c) $(p \to q) \wedge (q \to \neg p)$

Problem 3. **(8×3 points)**
Determine whether the following propositional statements are satisfiable. If a propositional statement is not satisfiable, use equivalences to show it (state the names of the equivalences used in your proof). If a propositional statement is satisfiable, provide an assignment of values to the variables.

(a) $\neg(p \vee r) \wedge (p \wedge r) \wedge s$

(b) $(a \vee b \vee c \vee \neg d) \wedge (\neg a \vee \neg b \vee \neg c \vee d)$

(c) $(p \to s) \wedge (s \to p)$

Problem 4. **(8×3 points)**
State whether the following statements are equivalent. If so, explain why. If not, explain why not using a counterexample. [Hint: If p and q are equivalent, then $p \leftrightarrow q$.]

(a) $\exists x(P(x) \wedge Q(x))$ and $\exists x P(x) \wedge \exists x Q(x)$

(b) $\forall x(P(x) \vee Q(x))$ and $\forall x P(x) \vee \forall x Q(x)$

(c) $\exists x \forall y(P(y) \leftrightarrow x = y)$ and $\exists x(P(x) \wedge \forall y(P(y) \to x = y))$

Problem 5. (7 $\times$ 2 points)
Let the universe be the set of all integers. Let $P(x)$ represent "x is negative" and $Q(x)$ represent "x is greater than 5." For each of the following formal logic expressions, determine if it correctly expresses the statement "no negative number is greater than 5," and justify your answer.

(a) $\neg\exists x\,(P(x) \rightarrow Q(x))$

(b) $\forall x\,(P(x) \rightarrow \neg Q(x))$

Problem 6. (14 points)
Rewrite $\neg\exists x\exists y\forall z(P(x) \vee \neg Q(y,z) \wedge R(z))$ such that the negation symbols immediately precede the predicates. Show your steps to receive full credit.
Note: $\wedge$ has higher precedence than $\vee$